

Buried Sphere Documentation

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Forward model

We model the vertical gravity anomaly at the surface produced by a buried sphere of depth z (to center), radius R , and density contrast $\Delta\rho$. For a station at horizontal radius $r = \sqrt{x^2 + y^2}$,

$$g(r; z, R, \Delta\rho) = G \left(\frac{4\pi}{3} R^3 \Delta\rho \right) \frac{z}{(r^2 + z^2)^{3/2}}, \quad (1)$$

with G the gravitational constant. This is nonlinear in z and linear in the *amplitude* $R^3 \Delta\rho$.

Analytical Jacobian

Let $A = G \frac{4\pi}{3}$ and $B(r, z) = \frac{z}{(r^2 + z^2)^{3/2}}$. Then $g = A R^3 \Delta\rho B$ and the Fréchet derivatives for a station at radius r are

$$\frac{\partial g}{\partial z} = A R^3 \Delta\rho \frac{\partial B}{\partial z}, \quad \frac{\partial B}{\partial z} = \frac{r^2 - 2z^2}{(r^2 + z^2)^{5/2}}, \quad (2)$$

$$\frac{\partial g}{\partial R} = 3 A R^2 \Delta\rho B = 3 A R^2 \Delta\rho \frac{z}{(r^2 + z^2)^{3/2}}, \quad (3)$$

$$\frac{\partial g}{\partial \Delta\rho} = A R^3 B = A R^3 \frac{z}{(r^2 + z^2)^{3/2}}. \quad (4)$$

For a dataset with stations $\{r_i\}_{i=1}^N$, the Jacobian J has rows $[\partial g_i / \partial z, \partial g_i / \partial R, \partial g_i / \partial \Delta\rho]$ evaluated at the current model.

Derivation of $\partial B / \partial z$. With $B = z(r^2 + z^2)^{-3/2}$,

$$\frac{\partial B}{\partial z} = (r^2 + z^2)^{-3/2} + z \left(-\frac{3}{2}\right) (r^2 + z^2)^{-5/2} (2z) \quad (5)$$

$$= \frac{1}{(r^2 + z^2)^{3/2}} - \frac{3z^2}{(r^2 + z^2)^{5/2}} = \frac{r^2 - 2z^2}{(r^2 + z^2)^{5/2}}. \quad (6)$$

Least-squares inversion (Gauss–Newton)

We seek $p = [z, R, \Delta\rho]^\top$ that minimizes a weighted least-squares objective

$$\Phi(p) = \|W_d (d - g(p))\|_2^2 + \lambda^2 \|W_m (p - p_{\text{ref}})\|_2^2, \quad (7)$$

where d are observations, W_d is the data-weighting (often $\sigma^{-1}I$), W_m is a model weighting (identity for 0th-order), p_{ref} is a reference model, and $\lambda \geq 0$ is the regularization (damping) parameter. Linearizing

$g(p) \approx g(p_0) + J(p - p_0)$ gives the normal equations for the increment Δp :

$$(J^\top W_d^\top W_d J + \lambda^2 W_m^\top W_m) \Delta p = J^\top W_d^\top W_d (d - g(p_0)) + \lambda^2 W_m^\top W_m (p_{\text{ref}} - p_0). \quad (8)$$

In the app, the default is $W_d = I$, $W_m = I$, $p_{\text{ref}} = p_0$, reducing to a Levenberg–Marquardt step

$$(J^\top J + \lambda^2 I) \Delta p = J^\top (d - g(p_0)). \quad (9)$$

Role of λ . Small λ approaches Gauss–Newton (fast, but can be unstable if $J^\top J$ is ill-conditioned). Large λ yields smaller, more stable steps (gradient-descent-like), helpful far from the solution or with strong trade-offs.

Uncertainty and resolution

After convergence to $\hat{p}$, approximate the model covariance by

$$\text{Cov}(p) \approx \sigma^2 (J^\top J + \lambda^2 I)^{-1}, \quad (10)$$

with σ^2 the residual variance $\|d - g(\hat{p})\|_2^2 / (N - M)$. For a parameter pair (indices i, j), extract the 2×2 submatrix and plot the 95% confidence ellipse using $\Delta\chi^2 \approx 5.991$ (2 DOF). The principal axes are given by the eigenpairs of the 2×2 covariance.

Units: SI vs mGal

The forward model uses SI throughout (m/s^2); many field datasets are displayed in mGal. Convert by $g_{\text{mGal}} = 10^5 g_{\text{SI}}$ and $\sigma_{\text{SI}} = 10^{-5} \sigma_{\text{mGal}}$.

Common diagnostics with the app

- **Pairwise vs global search:** slice minima need not coincide with the global minimum in 3–D. Differences highlight trade-offs and the effect of fixing the third parameter.
- **Inversion path:** the projected path on $(\Delta\rho, R)$, $(\Delta\rho, z)$, (R, z) reveals step size control by λ and curvature of Φ .
- **Noise and sampling:** increasing σ or coarsening station spacing widens the misfit basin and inflates uncertainty ellipses.

References (short list)

Further Reading:

- [1] R. Blakely, *Potential Theory in Gravity and Magnetic Applications*, Cambridge University Press, 1995.
- [2] W. Menke, *Geophysical Data Analysis: Discrete Inverse Theory*, 3rd ed., Academic Press, 2012.
- [3] R. C. Aster, B. Borchers, and C. H. Thurber, *Parameter Estimation and Inverse Problems*, 3rd ed., Elsevier, 2019.